A Neural Obstacle Avoidance PD End-effector Controller for Redundant Manipulators under Joint Constraints

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Abstract—Serial redundant robotic manipulators improve operational flexibility, especially in unstructured environments or tasks demanding high accuracy, outperforming conventional robot arms. Obstacle avoidance is a fundamental capability for robots to function efficiently in workspaces. This paper proposed a novel constrained obstacle avoidance approach to ensure effective collision avoidance as well as accurate end-effector trajectory tracking for redundant robotic manipulators under joint position constraints, without employing inverse kinematics solutions. In the proposed control structure, the first layer accomplishes obstacle avoidance, accounts for physical limitations, and effectively resolves the inverse kinematics problem. The second layer is employed to control the robot arm to achieve objectives established in the initial stage. By applying Lyapunov theory, the stability of the closed-loop system is clearly demonstrated under a given condition. The effectiveness of the proposed controller is verified on a three-degree-of-freedom planar robotic manipulator via intensive simulation results.

Index Terms—obstacle avoidance, redundant robotic manipulators, PD controller, end-effector controller, neural network.

I. Introduction

With the continuous progression of robotics, redundant robotic manipulators are increasingly endowed with higher degrees of freedom and superior dexterity, enabling them to execute complex tasks with greater precision and versatility [1], [2]. However, this redundancy also introduces additional control challenges, particularly when the manipulator must operate within confined and unstructured environments, avoid obstacles, and respect physical constraints [3], [4]. Achieving accurate end-effector trajectory tracking under such conditions, without relying on inverse kinematics, remains an active area of research in advanced robot control.

Obstacle avoidance and trajectory tracking for redundant robotic manipulators have been widely studied over the past decade due to their essential roles in safe and precise robot operation [5], [6], [7]. Methods based on task-priority redundancy resolution in joint space and task-space have been employed to handle simultaneous tasks [8], [9], [10]. Specifically, optimization-based techniques and artificial potential fields have been proposed for collision-free path planning while maintaining end-effector accuracy [11], [12]. The path planning method with obstacle avoidance was presented to assist manipulators to avoid obstacles and complete the tasks successfully in dynamic environment. However, many of approaches relied heavily on inverse kinematics, which can result in computational complexity and potential singularity problems in real-time applications [13], [14]. More recent works have aimed to perform direct end-effector control without solving inverse kinematics by employing null-space techniques to compute joint velocity control laws, yielding impressive control performance [5], [6], [15]. Such control strategies offer practical advantages in real-world applications, as they eliminate the need to explicitly solve the inverse kinematics problem.

Imposing joint constraints is essential to guarantee the operational safety and structural reliability of robotic manipulators. By using a new reward function designed to comply with safety constraints, a reinforcement learning-based reactive obstacle avoidance method can effectively coped with joint singularity and physical bounds automatically while ensuring the high quality control performance [16]. In [17], a hierarchical quadratic programming-prescribed performance control based control method was introduced for redundant wheeled mobile manipulators. This control strategy was capable of ensuring accurate trajectory tracking while simultaneously satisfying physical constraints and obstacle avoidance requirements during real-world operation. Besides that, robotic systems operating in uncertain environments must robustly handle both internal and external disturbances. To this end, disturbance observers, sliding mode observers, and adaptive neural networks have been explored for estimating unknown perturbations [18], [19]. Particularly, radial basis function (RBF) neural networks and recurrent neural networks have shown significant promise in modeling nonlinear and time-varying disturbances due to their strong function approximation capabilities [20], [21], [22]. Several studies have successfully combined neural networks with feedback controllers, including PD and sliding mode controllers, to compensate for torque disturbances and enhance control

robustness [20], [21]. Nonetheless, few existing works on obstacle avoidance for redundant manipulators incorporate such controllers under simultaneous task-space operation and joint angle constraints.

This paper proposes a new two-level neural obstacle avoidance PD control framework for redundant robotic manipulators in the task space, explicitly accounting for joint-space limitations. The primary contributions of this work are outlined as follows:

1) A high-level sub-controller is developed to achieve task-space control and obstacle avoidance objectives by appropriately generating desired joint velocities while respecting joint constraints. The control law is formulated based on the integration of the Jacobian analysis of the manipulator, an avoidance obstacle mechanism and the transformation function [23].
2) In the low-level sub controller, to effectively drive the system states follow well the referenced ones, a conventional PD scheme is designed. To compensate for internal and external disturbances in the manipulator dynamics, an extended Radial Basis Function (RBF) neural network signal is incorporated under a novel learning rule.
3) The stability of the closed-loop control system in task space is rigorously analyzed based on Lyapunov theory. Extensive comparative simulation results have confirmed the effectiveness of the proposed control strategy, which is particularly suitable for planar redundant robotic systems affected by internal and external disturbances.

The remainder of this paper is structured as follows. Section II presents the dynamic model of the system and formulates the control problem. Section III details the proposed control methodology along with the corresponding theoretical analysis. The performance of the developed controller is validated and compared with other control techniques in Section IV. Finally, Section V provides concluding remarks.

II. System Modelling and Problem Statements

The dynamics of a general n-DoF robotic system can be formulated using the Lagrangian approach in the following form [23]:

$$\mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{G}(\boldsymbol{\theta}) + \boldsymbol{\tau}_{\mathbf{f}} + \boldsymbol{\tau}_{\mathbf{dist}} = \boldsymbol{\tau} \tag{1}$$

where $\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}, \ddot{\boldsymbol{\theta}} \in \mathbf{R}^{n\times 1}$ are vectors of the joint angle, angular velocity and acceleration, respectively, $\mathbf{M}(\boldsymbol{\theta}) \in \mathbf{R}^{n\times n}$ is a symmetric positive definite mass matrix, $\mathbf{C}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) \in \mathbf{R}^{n\times n}$ is a vector representing the centrifugal-Coriolis moment, $\mathbf{G}(\boldsymbol{\theta}) \in \mathbf{R}^{n\times 1}$ is the gravity term, $\boldsymbol{\tau}_{\mathbf{f}} \in \mathbf{R}^{n\times 1}$ refers to frictional torque, $\boldsymbol{\tau}_{\mathbf{dist}} \in \mathbf{R}^{n\times 1}$ corresponds to external disturbances, and $\boldsymbol{\tau} \in \mathbf{R}^{n\times 1}$ is the control signal.

Remark 1: Indeed, the joint positions of the robotic manipulator are restricted to a finite range during operation.

$$\underline{\boldsymbol{\theta}} < \boldsymbol{\theta} < \overline{\boldsymbol{\theta}} \tag{2}$$

where $\underline{\boldsymbol{\theta}} \triangleq [\underline{\theta}_1, \underline{\theta}_2, ..., \underline{\theta}_n]^T$ and $\overline{\boldsymbol{\theta}} \triangleq [\overline{\theta}_1, \overline{\theta}_2, ..., \overline{\theta}_n]^T$ represent the lower and upper bounds of the joint position, respectively. In practice, surpassing these physical constraints can lead to potential risks affecting the control system

Let $(\mathbf{x_i})$ denote a position of the i-th joint of the robotic manipulators in the Cartesian coordinate system, where i ranges from 2 to n, and $\mathbf{x}$ represents a position of the end-effector. For a given robot, $(\mathbf{x_i})$ can be readily computed through forward kinematics, denoted as $(\mathbf{f_i})$, as expressed below.

$$\mathbf{x_i} = \mathbf{f_i}(\boldsymbol{\theta}(t)) \tag{3}$$

then $\dot{\mathbf{x}}_\mathbf{i} = \dot{\mathbf{f}}_\mathbf{i}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}(t) = \mathbf{J}_i\mathbf{K}_i\dot{\boldsymbol{\theta}}(t)$ where $\mathbf{J}_i$ is the Jacobian matrix of the i-th joint and $\mathbf{K}_i$ is the gain matrix used to select appropriate joint angles. To simplify the implementation, in this paper we have the following assumptions:

Assumption 1: The redundant manipulator is restricted to planar motion, and obstacles are modeled as circles by a center and radius $(\mathbf{R_1})$. To ensure safe operation, a minimum safety distance $(\mathbf{R_2})$ from the obstacle center to joints of the robotic manipulator is enforced, where $(\mathbf{R_2})$ is greater than $(\mathbf{R_1})$.

Assumption 2: Only the joints of the arm avoid obstacles, not the links of the robotic manipulators.

Let the position and velocity of the obstacle's center be represented by $\mathbf{x_{obs}}$ and $\dot{\mathbf{x}}_\mathbf{obs}$, respectively. Based on the assumption 2, we define the distance from the ith joint to the center of the obstacle as follows:

$$\mathbf{d_{iobs}} = (\mathbf{x_i} - \mathbf{x_{obs}})^T(\mathbf{x_i} - \mathbf{x_{obs}}) \tag{4}$$

Remark 2: Considering the desired trajectory $(\mathbf{x_d})$ and velocity $(\dot{\mathbf{x}}_\mathbf{d})$ of the end-effector, this paper focuses on proposing a practically applicable control method to generate a suitable control signal $(\boldsymbol{\tau})$, ensuring that the end-effector of the robot can effectively track the reference trajectory and avoid the obstacle under physical constraints. Although inverse kinematics (IK) is generally applied to compute the joint angles from the desired end-effector trajectory, the complexity of solving the IK problem often requires the use of specialized solution methods. To ensure generality, we convert the dynamic model (1) into its task-space representation [23], as follows:.

$$\dot{\mathbf{x}} = \mathbf{J}\dot{\theta} \tag{5}$$

where $\dot{\mathbf{x}}$ is a linear velocity of the end-effector of the robotic manipulators and $\mathbf{J}$ is a Jacobian matrix.

Remark 3: In practice, precise identification of the dynamic model (1) is challenging, as system parameters may vary with operating conditions. Thus, the proposed controller is designed to be model-free, adaptive, and robust.

III. Neural Obstacle Avoidance PD End-effector Controller

This section introduces a novel controller designed to accomplish the specified control objectives. The controller

employs a high-level control framework integrated with a neural-PD controller to ensure that the end-effector tracks the desired trajectories while effectively avoiding obstacles. System stability is analyzed using an extended Lyapunov approach.

A. High-Level Control Structure

The control strategy at this level aims to generate the desired position $\boldsymbol{\theta_d}$ based on the reference end-effector trajectory, location and size of the obstacle and the given forward kinematics. To achieve this, a novel transformation function ($\mathbf{g}$) is employed to map the joint position from a finite interval into an infinite domain, as expressed below:

$$\theta_i(t) = \mathbf{g}(q_i(t)) = \frac{\overline{\theta}_i - \underline{\theta}_i}{1 + e^{-\lambda q_i}} + \underline{\theta}_i \tag{6}$$

where $\mathbf{q} = [q_1, q_2, ..., q_n]^T$ is a virtual joint position and λ represents the slope of the function. It should be noted that as q_i ranges from negative infinity to positive infinity, the joint position stays within its permitted bounds.

We define the primary control objective as follows:

$$\mathbf{e} = \mathbf{x} - \mathbf{x_d} \tag{7}$$

A desired virtual angular velocity ($\dot{\mathbf{q}}_\mathbf{d} \triangleq [\dot{q}_{d1}, \dot{q}_{d2}, ..., \dot{q}_{dn}]^T$) is formulated based on the following rule:

$$\dot{\mathbf{q}}_\mathbf{d} = \mathbf{K}{\mathbf{J_g}}^{-1}(-\boldsymbol{\eta}\mathbf{A}^T - \mathbf{A}^+\mathbf{B}) \tag{8}$$

where $\mathbf{K} = \text{diag}([k_{11}, k_{12}, ..., k_{1n}]) \in \mathbf{R}^{n\times n}$ is a positive-definite diagonal matrix and $\mathbf{J_g}(\mathbf{q}) \triangleq \text{diag}[\dot{\mathbf{g}}(q_1), \dot{\mathbf{g}}(q_2), ..., \dot{\mathbf{g}}(q_n)]$, $\boldsymbol{\eta}$ is a positive coefficient, $\mathbf{A}^+$ is a pseudo-inverse matrix of $\mathbf{A}$, and $\mathbf{A}, \mathbf{B}$ are defined as follows:

$$\begin{cases} \mathbf{d}_i \triangleq e^{-(\mathbf{d_{iobs}} - \mathbf{R}_1^2)} \\ \boldsymbol{\alpha} \triangleq \frac{1}{\boldsymbol{\beta}} e^{-(\mathbf{R}_2^2 - \mathbf{R}_1^2)} \\ \mathbf{a}_{ii} \triangleq \mathbf{a}_i \boldsymbol{\mu} \frac{e^{-\boldsymbol{\mu}(\mathbf{d_i} - \boldsymbol{\alpha})}}{(1 + e^{-\boldsymbol{\mu}(\mathbf{d_i} - \boldsymbol{\alpha})})^2} \\ \mathbf{A} \triangleq \boldsymbol{\gamma}\mathbf{e}^T\mathbf{J} - \sum_{i=2}^{n}[\mathbf{a}_{ii}\mathbf{d_i}(\mathbf{x_i} - \mathbf{x_{obs}})^\mathbf{T}\mathbf{J}_i\mathbf{K}_i] \\ \mathbf{B} \triangleq -\boldsymbol{\gamma}\mathbf{e}^T\dot{\mathbf{x}}_\mathbf{d} + \sum_{i=2}^{n}[\mathbf{a}_{ii}\mathbf{d_i}(\mathbf{x_i} - \mathbf{x_{obs}})^\mathbf{T}\dot{\mathbf{x}}_\mathbf{obs}] \end{cases} \tag{9}$$

Remark 4: As shown in (8), the high-level controller generates reference virtual angular velocity and corresponding position from the desired end-effector trajectory and the motion path of the obstacle under joint limits. Unlike conventional inverse kinematics [23], the control law in (8) inherently considers obstacle avoidance and joint constraints. The desired joint angles ($\boldsymbol{\theta_d}$) can be readily derived by incorporating (6).

B. A PD Feedback Control Strategy

The control law is formulated based on a standard proportional-derivative (PD) scheme, as shown below:

$$\tau = \mathbf{K_p}\mathbf{e}_\theta + \mathbf{K}_v\dot{\mathbf{e}}_\theta \tag{10}$$

where $\mathbf{e}_\theta \triangleq \boldsymbol{\theta} - \boldsymbol{\theta_d}$ is a vector of joint control error, $\dot{\mathbf{e}}_\theta \triangleq [\dot{e}_{\theta_1}, \dot{e}_{\theta_2}, ..., \dot{e}_{\theta_n}]^T$ is the time derivative of the joint control error vector, control gains are $\mathbf{K_p} = \text{diag}([k_{p1}, k_{p2}, ..., k_{pn}]) \in \mathbf{R}^{n\times n}$ and $\mathbf{K_v} = \text{diag}([k_{v1}, k_{v2}, ..., k_{vn}]) \in \mathbf{R}^{n\times n}$.

As shown in (8), (6) and (10), for each target position, the high-level controller computes the virtual joint reference while respecting joint limits and collision avoidance . The PD control scheme then drives the actual joint position to follow, ensuring the end-effector tracks the desired trajectory. As a result, the end-effector trajectory of the robotic manipulators—from the beginning to the end of the motion—remains within joint limits and is capable of obstacle avoidance, while maintaining high control performance.

C. A Supplementary Control Signal Generated by a Neural Network

The disturbance $\boldsymbol{\tau}_{dist}$ is estimated via a Radial Basis Function (RBF) network as follows:

$$\boldsymbol{\tau_{dist}} = \mathbf{J}^T\mathbf{w}^T\boldsymbol{\xi} + \boldsymbol{\delta} \tag{11}$$

where $\mathbf{w} = [w_1 w_2]$ and $\boldsymbol{\xi}$ are respectively the optimal weighting matrix and regression state vector, $\boldsymbol{\delta}$ is the modeling errors and external disturbances.

Remark 5: The term $\boldsymbol{\tau_{dist}}$ denotes internal and external disturbances affecting the manipulator dynamics. In practical robotic applications, the system can perform tasks only if the disturbance energy is finite. Accordingly, the following assumption is introduced::

Assumption 3: A bounded optimal weight matrix ($\mathbf{w}$) is assumed in the analysis.

Based on (11), control law (10) is enhanced by incorporating the intelligent term to ensure system robustness against disturbances, as follows:

$$\tau = \mathbf{K_p}\mathbf{e}_\theta + \mathbf{K}_v\dot{\mathbf{e}}_\theta + \mathbf{J}^T\widehat{\mathbf{w}}^T\boldsymbol{\xi} \tag{12}$$

where $\widehat{\mathbf{w}}$ is an estimated optimal weight matrix of the neural network. The training law of RBF neural network is designed as:

$$\dot{\hat{w}}_{i|i=1,2} = -\rho_i\hat{w}_i - \vartheta_i\mathbf{e}_i\boldsymbol{\xi} \tag{13}$$

where ρ_i is a positive leakage rate, ϑ_i are positive learning rates and $\mathbf{e}_i$ denotes the tracking error vector of the robot end-effector in the x and y directions.

As observed in (13), the control error signals are leveraged to perform online adaptation of the neural network's weight matrix, allowing the control errors to converge gradually toward minimal values.

D. Stability Analysis

This section presents the stability analysis of the proposed control scheme for robotic manipulator. We derived the following statement from the suggestions above.

Theorem 1: Given a dynamic equation of robotic manipulators (1) under joint constraints (2), and using the desired joint angles from the high-level sub controller (8), the neural PD control signal (12) integrated with the

nonlinear learning rules (13), the control error defined in (7) asymptotically converges to a neighborhood of zero from any initial condition, without violating the predefined physical constraints and obstacle ones by choosing appropriate control gains and learning coefficients.

Proof:

We propose a new Lyapunov function:

$$V = \frac{1}{2}\boldsymbol{\gamma}\mathbf{e}^{\mathbf{T}}\mathbf{e} + \sum_{i=2}^{n} \mathbf{a}_i \frac{1}{1+e^{-\boldsymbol{\mu}(\mathbf{d}_i-\boldsymbol{\alpha})}} \tag{14}$$

Differentiating the function (14) with respect to time yields:

$$\begin{aligned}\dot{V} =& \boldsymbol{\gamma}\mathbf{e}^{\mathbf{T}}(\mathbf{J}\dot{\boldsymbol{\theta}} - \dot{\mathbf{x}}_{\mathbf{d}}) \\ &- \sum_{i=2}^{n} \mathbf{a}_i\boldsymbol{\mu}\frac{\mathbf{e}^{-\boldsymbol{\mu}(\mathbf{d}_i-\boldsymbol{\alpha})}}{(\mathbf{1}+\mathbf{e}^{-\boldsymbol{\mu}(\mathbf{d}_i-\alpha)})^{\mathbf{2}}}\mathbf{d}_i(\mathbf{x}_i-\mathbf{x}_{\mathbf{obs}})^{\mathbf{T}}(\dot{\mathbf{x}}_i-\dot{\mathbf{x}}_{\mathbf{obs}})\end{aligned} \tag{15}$$

Utilizing the definition in (9), the equation (15) can be rewritten:

$$\begin{aligned}\dot{V} =& [\boldsymbol{\gamma}\mathbf{e}^{\mathbf{T}}\mathbf{J} - \sum_{i=2}^{n} \mathbf{a}_{\mathbf{ii}}\mathbf{d}_i(\mathbf{x}_i-\mathbf{x}_{\mathbf{obs}})^{\mathbf{T}}\mathbf{J}_i\mathbf{K}_i]\dot{\boldsymbol{\theta}} \\ &+ [-\boldsymbol{\gamma}\mathbf{e}^{\mathbf{T}}\dot{\mathbf{x}}_{\mathbf{d}} + \sum_{i=2}^{n} \mathbf{a}_{\mathbf{ii}}\mathbf{d}_i(\mathbf{x}_i-\mathbf{x}_{\mathbf{obs}})^{\mathbf{T}}\dot{\mathbf{x}}_{\mathbf{obs}}]\end{aligned} \tag{16}$$

The dynamics in (16) simplify to:

$$\dot{V} = \mathbf{A}\dot{\boldsymbol{\theta}} + \mathbf{B} \tag{17}$$

We choose

$$\dot{\boldsymbol{\theta}}_{\mathbf{d}} = -\boldsymbol{\eta}\mathbf{A}^{T} - \mathbf{A}^{+}\mathbf{B} \tag{18}$$

Consequently, the control objective $\mathbf{e}$ is shown to be bounded. By combining (6) and (18), it can be seen that the proof of Theorem 1 is completed .

Block diagram of the proposed controller is described in Fig. 1

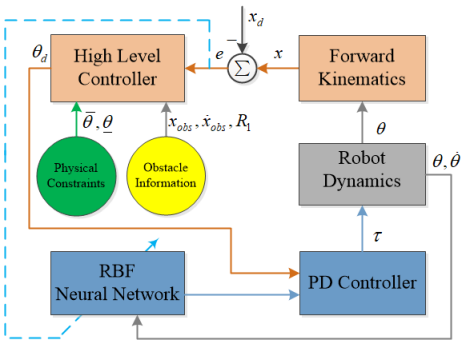


Fig. 1. Control scheme of the robotic system.

IV. Validation Results

In this section, the effectiveness of the proposed algorithm is assessed through simulations. The closed-loop control strategy is applied to a three-link planar redundant robotic manipulator, as depicted in Fig. 2.The structure of the manipulator consists of three rigid links with lengths of $(\mathbf{l_1}, \mathbf{l_2}, \mathbf{l_3})$ .The masses $(\mathbf{m_1}, \mathbf{m_2}, \mathbf{m_3})$ are placed at the end position of each link. The frictional torque produced by viscous phenomena is proportional to the velocity of the joint motion and viscous constants $(\mathbf{a_1}, \mathbf{a_2}, \mathbf{a_3})$.

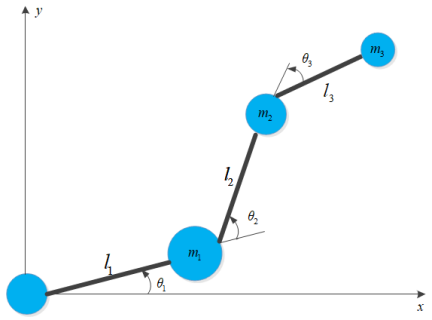


Fig. 2. The testing robot.

The joint angle constraints for the robot were specified as follows:

$$\begin{cases} \underline{\theta}_1 = -\pi(\text{rad}); \bar{\theta}_1 = \pi(\text{rad}) \\ \underline{\theta}_2 = -3\frac{\pi}{4}(\text{rad}); \bar{\theta}_2 = 3\frac{\pi}{4}(\text{rad}) \\ \underline{\theta}_3 = -4\frac{\pi}{5}(\text{rad}); \bar{\theta}_3 = 4\frac{\pi}{5}(\text{rad}) \end{cases}$$

An RBF neural network was employed to approximate the disturbance term with high accuracy. The network consisted of six input neurons, corresponding to joint positions (θ_i) and joint velocities $(\dot{\theta}_i)$. Each input was encoded using three Gaussian basis functions, resulting in a hidden layer with 3^6 neurons. Two output neurons were used to generate the network output, and all weight vectors were initialized to zero. The hidden neuron centers were uniformly distributed across the input domain using a grid-based method. The widths of the Gaussian functions were determined by the average inter-center distance. The control and learning gains were appropriately chosen to ensure the stability of the closed-loop system.

To thoroughly evaluate the adaptability and robustness of the proposed control algorithm (denoted as PRO), its simulation performance was compared under various operating conditions with that of the neural obstacle avoidance PD end-effector controller without considering joint position constraints (referred to as NC), and the non-adaptive obstacle avoidance PD end-effector controller which accounts for physical bounds (called as NA).

The performance of the proposed controllers was validated through two simulation scenarios involving the redundant robotic manipulator. The first scenario examined the controller under no external load, while a payload applied to the robot arm was introduced to evaluate its performance under load conditions in the

second simulation. The parameters of the redundant robotic manipulator and the controllers were presented in Table I. In both cases, the effects of internal and external disturbances were incorporated into the model as additive terms influencing the actuator torques. In addition, process-level disturbances were characterized by white noise, as illustrated in Fig. 3.

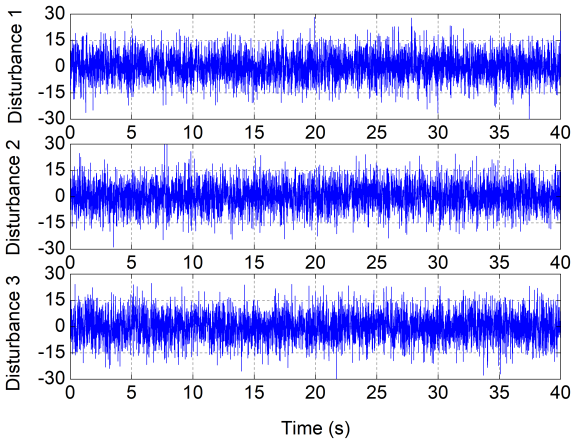


Fig. 3. Disturbances affected the robotic manipulator in both validation scenarios.

Control results of the first simulation case are illustrated in Figs. 4 and 5, following the implementation of both the proposed and comparative controllers on the testing robot.

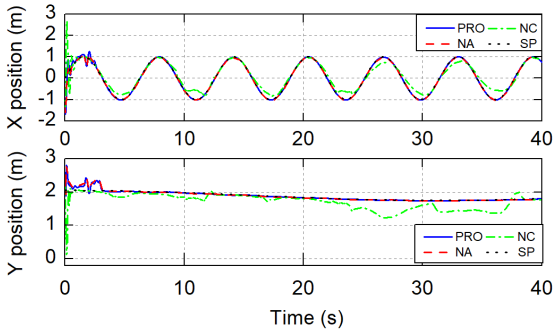


Fig. 4. The position of the end-effector in the first simulation.

As shown in these figures, the performance characteristics of the controllers can be distinctly observed through the variations in the control data. As observed in Fig. 5, the PD controller without physical constraints exhibited the largest control error. Generally, control performance of the proposed controller and the PD controller that is not integrated neural network were the same. However, a more detailed observation of this figure revealed that the neural-based PD controller achieved superior tracking accuracy. This superior performance was attributed to the neural network's ability to estimate the internal and external

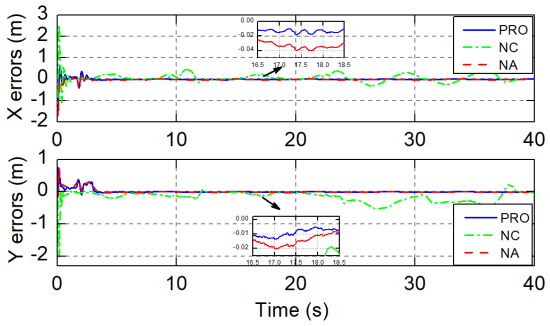


Fig. 5. Control errors of the controllers in the first simulation.

disturbances acting on the robotic arm and to generate appropriate torques that compensate for these effects. The results illustrated in Figs. 6 through 9 indicated that the proposed controller yielded substantially enhanced steady-state performance in comparison with the alternative methods. This improvement originated from a key feature of the proposed control scheme. The high-level sub controller was designed to generate the desired joint angles under physical joint constraints, enabling the manipulator to track the referenced trajectory while avoiding collisions, based on the known obstacle positions and velocities. Then, the low-level controller, the conventional PD controller integrated with the RBF neural network,was responsible for driving the joint angles of the manipulator to track the reference signals generated by the high-level ones. As depicted in Fig. 6, both the proposed controller and the non-adaptive PD controller were able to compute the desired joint angles of the manipulator while respecting the predefined physical constraints. In contrast, the unconstrained controller generated joint trajectories that violated the physical joint limits. This clearly demonstrated the superiority of the joint-constraint based algorithm employed in the proposed approach. The training process of the nonlinear RBF neural network was described in Fig. 7. The variation of the weight vector from zero to specific values, governed by the adaptive update law, enabled the network to effectively learn the internal and external disturbances acting on the system. This learning process contributed to minimizing the end-effector position error, thereby improving overall tracking performance. Figure 8 shown the Euclidean distances from the second and third joints of the redundant robot to the center of the obstacle. Figure 9 illustrated the joint torques, with peak values during the transient phase, enabling rapid system response and fast stabilization. Notably, at around 7.5 and 14 seconds—when the manipulator reached its closest proximity to the obstacle—sharp variations in the control signals occurred due to the activation of the obstacle avoidance component in the high-level controller. This term enforced repulsive actions based on proximity, allowing the system to maintain high

tracking accuracy while respecting safety constraints.

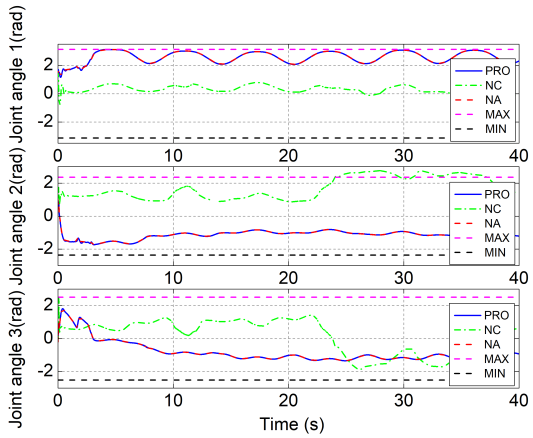


Fig. 6. Desired joint angles generated by high-level controllers in the first simulation.

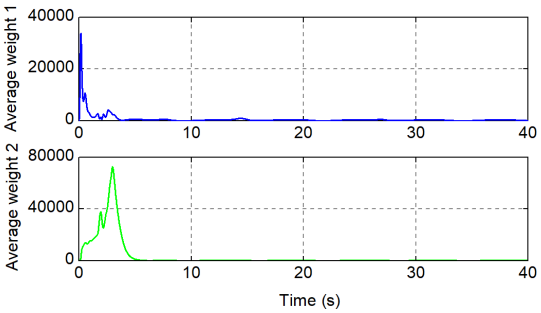


Fig. 7. Estimation results of RBF neural network.

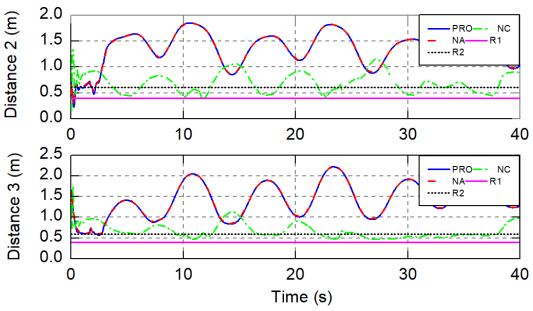


Fig. 8. The Euclidean distances from joints to the center of the obstacle.

To further evaluate the performance of the controller, the closed-loop system is tested under load conditions. The external load is assumed to act uniformly on all joints of the robotic manipulator, with the masses of the joints specified in Table I. The task-space control errors of the controllers are depicted in Fig. 10.

Figure 10 shown that the proposed controller, enhanced with an adaptive mechanism, maintained superior perfor-

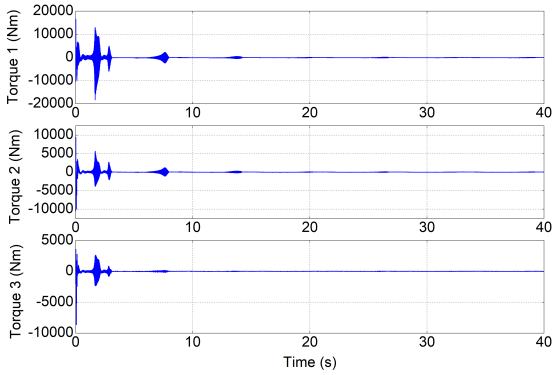


Fig. 9. Control torques generated by the proposed controller.

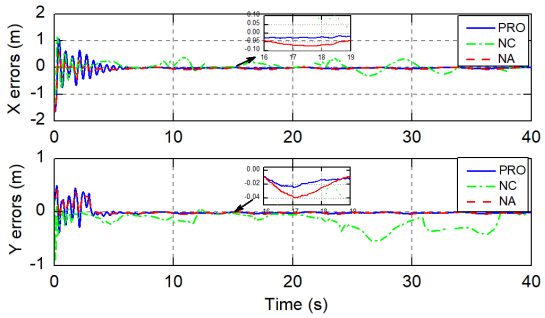


Fig. 10. Control errors of the controllers in the loaded condition.

mance in the steady- state period under load condition compared to the other controllers. Therefore, the ability of the suggested controller to accomplish the task under joint position constraints, as well as in the presence of internal and external disturbances, has been verified through simulation results. In simulations, the end-effector position of the robotic manipulator is typically determined using forward kinematics equations. While this approach is effective in an ideal environment, its direct application to real-world systems often results in inaccuracies caused by mechanical imperfections and structural errors. To address these limitations, a 3D camera is integrated into the system to directly measure the end-effector position in physical space, thereby providing corrective feedback and significantly enhancing the overall accuracy and reliability of the control process.

V. Conclusions

This paper proposes a novel two-level task-space obstacle avoidance control strategy for redundant robotic manipulators that enables accurate end-effector trajectory tracking and effective obstacle avoidance under joint angle constraints, without requiring the resolution of conventional inverse kinematics. The nonlinear control signal is derived by combining a joint-velocity generation method

TABLE I
The Parameters' Values of the Redundant Robot and the Controllers

Symbol	Name	Values
Parameters of the redundant robot		
$\mathbf{m_1}$, $\mathbf{m_2}$, $\mathbf{m_3}$	joint masses (kg) in case 1 and 2	5, 3, 2 and 10, 6, 4
$\mathbf{a_1}$, $\mathbf{a_2}$, $\mathbf{a_3}$	viscous friction coefficients	20, 20, 20
$\mathbf{l_1}$, $\mathbf{l_2}$, $\mathbf{l_3}$	link lengths (m)	1, 1, 1
NC controller		
$\mathbf{K_p}, \mathbf{K_v}$	control gains in case 1 and 2	$\mathrm{diag}(-[6200; 9300; 7440])$, $\mathrm{diag}(-[200; 100; 100])$
NA controller		
$\mathbf{K}_p, \mathbf{K}_v$	control gains in case 1 and 2	$\mathrm{diag}(-[7000; 10500; 8400])$, $\mathrm{diag}(-[200; 100; 100])$
Proposed controller		
$\mathbf{K}_p, \mathbf{K}_v$, $\mathbf{K}$	control gains in case 1 and 2	$\mathrm{diag}(-[7000; 10500; 8400])$, $\mathrm{diag}(-[200; 100; 100])$, $\mathrm{diag}([1.5; 1; 1.5])$
η, λ,β,μ, γ, $\mathbf{a}_1$ and $\mathbf{a}_2$	control terms in case 1 and 2	20, 1, 0.8, 10, 1, 7 and 7
$\rho_{i\|i=1..2}$ and $\vartheta_{i\|i=1..2}$	learning coefficients	$\mathrm{diag}([0.8; 0.8])$and $\mathrm{diag}([1000; 1000])$,

that respects physical limitations with a conventional PD control scheme augmented by a Radial Basis Function (RBF) neural network. Convergence of the control performance is guaranteed through the appropriate design of learning rates and control gains. The integration of the newly developed control approach and the adaptive learning capability of the neural network improves robustness and ensures high-precision control in the presence of system uncertainties and unknown disturbances. Despite its promising performance, future work will focus on refining the control law to enhance tracking accuracy for diverse reference signals in high-degree-of-freedom robotic systems, as well as implementing the scheme in real-time applications.

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